

AI Swarm Drones Find Survivors 3- 4x Faster Than Traditional SAR

10-18
MIN/MI²78-85%
DETECTION12%
FALSE
ALARMS\$12K
SORTIE50Hz
OFFBOARD

SUBSYSTEM A //

GNC

**SUTRA-
FSD GNC &
3D ORCA**SOLVES PROBLEM
VOID:Canopy GPS
Multi-Path
Drift &
CrashesSOLUTIONAL
MOAT:Decentralized
3D
velocity
obstacle
collision
avoidance
+ PX4
MicroXRCE
50Hz
offboard
setpoint
streaming.0.05s
Response
Latency0
Tower
Dependency-5dB
RESILIENT

SUBSYSTEM B //

COMMS

**Deep
JSCC
Neural
Video**SOLVES PROBLEM
VOID:Mountain
Ravine RF
Ridge
BlackoutSOLUTIONAL
MOAT:Differentiable
joint
source-
channel
coding
yielding
zero
digital
cliff and
graceful
video
degradation
down to
-5dB.3.59cm
ACCURACY

SUBSYSTEM C //

VISION

**3D DEM AI
Geolocation**SOLVES PROBLEM
VOID:35% False
Alarms & Flat
GPS DriftSOLUTIONAL
MOAT:Sub-15ms
edge
YOLOv8-
Pose with
real-time
WGS84
elevation
raycasting,
eliminating
15-30m
calculation
errors.<15ms
Inference
Time3.59cm
Geo
Error1-2
OPERATORS

SUBSYSTEM D //

C2 GCS

**Pegasus
3D
WebGPU
GCS**SOLVES PROBLEM
VOID:15-25 Ground
Crew
Cognitive
OverloadSOLUTIONAL
MOAT:Offline
tactical
3D
digital
twin +
ATAK
plugin,
empowering
1-2
operators
to
orchestrate
full
drone
swarms.

-5dB
SNR
Threshold

0
Cliff
Degradation

**1-2
Ops**
Crew
Size

100%
Offline
ATAK

DECENTRALIZED 50Hz

ELEVATION.JETSON